**THE PRESIDENT'S OFFICE**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**
**KILIMANJARO REGIONAL COMMISSIONER'S OFFICE**

**FORM FOUR MOCK EXAMINATION**

**032 CHEMISTRY 1**
**(For Both School and Private Candidates)**

**Time: 3:00 Hours** **May. 2024**

MARKING SCHEME

**SECTION A: (16 marks )**

| i. | ii. | iii. | iv. | v. | vi. | vii. | viii. | ix. | x. |
|---|---|---|---|---|---|---|---|---|---|
| D | D | D | B | B | D | C | A | C | C |

*10@01mark*

| LIST A | i. | ii. | iii. | iv. | v. | vi. |
|---|---|---|---|---|---|---|
| LIST B | H | I | F/A | B | C | L |

*06@01mark*

**SECTION B** (54 marks)

3. (a) No. The room does not qualify to be used as a chemistry laboratory. Because a good laboratory must have**;** ***(01 mark).***

(i) Large window for enough ventilation.
(ii) Water supply system for proper conduction of experiments and cleanliness.
(iii) Drainage system for proper waste disposal.
(iv) Door open outward for easy escape in case of accident. ***(04 @ o1marks).***

Other -Electricity supply to facilitate heating and lighting.

-Gas supply to facilitate heating of chemical substances.

(b) A suspension will have visible insoluble particles which will settle on being left to stand while a solution is clear and no particles will settle on being left to stand. *02marks @ 01mark*

(c) Oxygen gas is used in both
Both they form oxides. *02@01mark*

04. (a) $S_{(s)}+Fe_{(s)}$ **----heat---->**$FeS_{(s)}$ ***(03 marks)***

(b) .The mixture continue glowing even after heating has been stopped because the reaction is exothermic (giving heat out). ***(03 marks).***

(c) The product formed in (a) cannot be attracted by a magnet because the magnetic property of iron is lost when the compound is formed. ***(03 marks)***

5. answers

i. Experimentation and observation.

(ii) Problem identification.
(iii) Drawing conclusion

(iv) Formulation of hypothesis. ***(0.5@6=3marks)***

(v) Data collection and recording

(vi) Data analysis and interpretation.

( b) Importance of scientific procedure.***(any three)***

***i.*** For carrying out experiments.
Experiments are scientific test that are done systematically to study what happens and to gain new knowledge.

***ii.*** Applied in project work .
A project is a planned piece of work that involves careful study of a subject matter or a problem over a period of time to find information on the subject or problem.

***iii.*** Applied in field study (Field work) A field study involves doing practical work in real environment in order to find answer to problems and to test hypothesis. ***(02@3=6marks)***

6. (a) Conductivity decrease across period 3 elements because on the left hand side the elements are purely metallic with free electrons. As you more across the period, elements become more molecular where electrons are shared and no free electrons are available in their structure. *03marks*

(b) Moles of copper deposited, $n = \frac{\text{mass (g)}}{\text{Molar mass (glmol)}}$ *01mark*

$n = \frac{0.32g}{64glmol} = 0.005$ moles *01mark*

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

From the equation, 1mole of copper is deposited by 2faradays
0.005mol of copper are deposited by X faradays

$\frac{2F \times 0.005mol}{1mol} = 0.01F$

Then, Quantity of electricity (coulombs) required = 0.01F x 96500C/F
= 965C *01mark*

OR
1F = 96500C
0.01F = x

$X = \frac{0.01F \times 96500C}{1F} = 965C$

(c) - concentration
- Pressure
- Temperature *any three well explained. 03@01mark*
- Catalyst

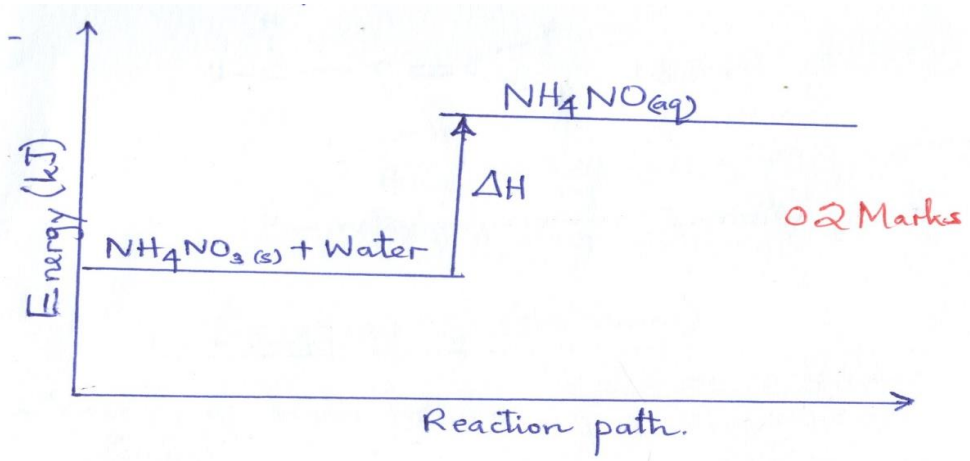
7. (a)

***(01@6 = 6marks)***

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| III | VI | IV | II | V | I |

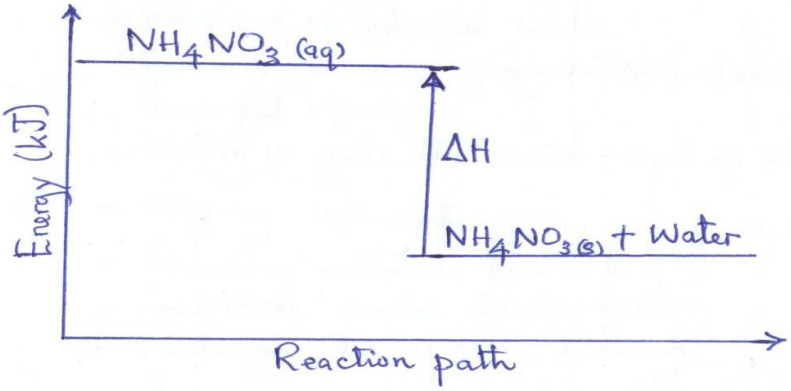
(B) It contains sulphates of magnesium and calcium. These form scum with soap leading to soap wastage. *01marks*

(b) Endothermic process.




02 Marks

OR.




08. (a) Carbon have been given special attention in organic chemistry rather than other elements because;
i. Has ability to form stable bonds with many elements

(ii) Can form bonds with a long chain of other carbon atoms (catenation) ***(01.5@2=03 marks)***

(iii) Forms large number of covalent bonds (form four bonds) ***(Any two reasons)***

(b)(i) A- Alkane. B- Alkene. C- Alcohol (alkanol). D- Carboxylic acid. (***0.5@4=02 marks).***

(ii) Hydrogenation reaction. ***(02 Mark).***

(iii) No any reaction can take place since methane reacts with halogen only in presence of sunlight***.(02 Mark)***

**SECTION C (30 marks)**

Answer two (2) questions in this section

9. Meaning of water pollution *introduction 1.5marks*

Points:

- Treatment of sewage before it is discharged into the water bodies.
- Recycling of industrial wastes.
- Reducing the use of agricultural chemicals especially those with harmful effect.
- Using organic fertilizers such as animal manure.
- Stopping deforestation.
- Reducing pollution from oil.
- Controlling coastal development.
- Reducing the use of mercury. *any six points 12@02marks*
- Cleaning up existing and abandoned mines.
- Cleaning up chemical pollutants.

Conclusion. *1.5marks*

10. Negative impacts of metal extraction to the environment and life of living things ***(any five points)***
-Health problems caused by toxic metals like mercury

-Air pollution which lead to global warming

- Loss of biodiversity.

-Acid rain due to emission of harmful gases eg. Carbon dioxide and sulphur dioxide.

-Make environment look ugly due to accumulation of solid waste and creation of pits.

***N:B Any relevant point should be awarded. (Introduction 01 marks , points 02.5@5=12.5marks , relevant conclusion 01.5marks)***

11. (a) Safety measure is all possible things that an individual should do to protect him / her form danger. *2.5marks*
(b)
(i) A chemistry laboratory should have safety measure to minimize or injuries and exposure to harmful substance. *2.5marks*
(ii) It is necessary to familiarize yourself with laboratory apparatus so that we can know the uses and how to handle when in use. *2.5marks*
(iii) Chemicals in the laboratory must be label and stoppered so as to avoid accidental use of wrong substance. *2.5marks*
(iv) All people working in the laboratory should wear appropriate protective clothes so as to minimize exposure to hazards. *2.5marks*
(v) Refrigerators and freezers used in the laboratory must be labeled for chemical used only so as avoid contamination with other substances. *2.5marks*